

Lecture 03: Quantum Physics & Schrödinger's Equation

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MAT SCI 120 — Winter 2026

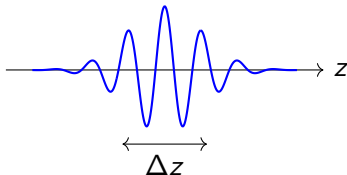
The wave-particle duality problem

- ▶ Electrons diffract like waves (Davisson-Germer) but hit detectors as particles
- ▶ Classical mechanics cannot explain interference patterns from single electrons
- ▶ Need new framework: neither pure wave nor pure particle description works
- ▶ We need to explain how electrons behave as particles (localized) or waves (delocalized)

How to localize waves? Building a wave packet

Superposition principle:

- ▶ Combine waves with wavenumbers spanning range Δk around central k_0
- ▶ Waves add **constructively** in localized region, **cancel** elsewhere
- ▶ Narrower spatial extent Δz requires broader spread Δk



Fundamental trade-off: Localization in space $\leftrightarrow$ spread in wavenumber

The space-wavenumber uncertainty

Fourier analysis of wave packets reveals a fundamental constraint:

- ▶ Product of spatial width and wavenumber spread is constant
- ▶ We cannot make both Δz and Δk arbitrarily small simultaneously

Spatial-Wavenumber Uncertainty

$$\Delta k \cdot \Delta z = 2\pi \quad (1)$$

Interpretation:

- ▶ Small Δz (well-localized) $\Rightarrow$ large Δk (many wavenumbers)
- ▶ Small Δk (well-known momentum) $\Rightarrow$ large Δz (spread out)

Uncertainty in momentum

Connecting wave and particle descriptions:

The de Broglie relation links wavenumber to momentum:

de Broglie's connection of momentum and wavenumber

$$p = \hbar k = \frac{h}{2\pi} \cdot \frac{2\pi}{\lambda} = \frac{h}{\lambda} \quad (2)$$

Key consequence:

- Uncertainty in k implies uncertainty in momentum:

$$\Delta p = \hbar \Delta k$$

- Substitute into $\Delta k \cdot \Delta z = 2\pi$ to obtain position-momentum relation

Heisenberg uncertainty principle

Starting point: $\Delta k \cdot \Delta z = 2\pi$ and $\Delta p = \hbar \Delta k$

Derivation:

$$\Delta k \cdot \Delta z = 2\pi \quad (3)$$

$$(\hbar \Delta k) \cdot \Delta z = 2\pi \hbar = h \quad (4)$$

$$\Delta p \cdot \Delta z = h \quad (5)$$

Heisenberg uncertainty principle

$$\boxed{\Delta p \cdot \Delta z = h} \quad (6)$$

Crucial insight: This is a **fundamental limit** on simultaneous knowledge of position and momentum—not a measurement limitation, but an intrinsic property of quantum reality.

Uncertainty is a fundamental property of matter

The uncertainty principle is NOT due to:

- ▶ Imperfect instruments or imprecise measurements
- ▶ Technological limitations

It IS about the nature of wave-particle duality:

- ▶ A particle *does not have* definite p and z simultaneously
- ▶ Wave nature makes simultaneous sharp values of p and z *impossible*

Physical consequence: Explains atomic stability

- ▶ If electron collapsed onto nucleus: $\Delta z \rightarrow 0$
- ▶ Then $\Delta p \rightarrow \infty$ — kinetic energy explodes
- ▶ Atoms find stable size balancing localization vs. kinetic cost

Phase and group velocity

A wave packet contains many frequency components

- ▶ Individual wave crests move at **phase velocity** $v_p = \omega/k$
- ▶ The envelope (packet shape) moves at **group velocity**
- ▶ These are generally *different* speeds!

Central question:

Which velocity represents the actual electron motion?

- ▶ Answer determines how we connect wave description to particle dynamics

Phase velocity is the wave motion

Definition: Phase velocity describes constant-phase surface propagation

For a plane wave $\psi = e^{i(kz - \omega t)}$, constant phase requires:

$$kz - \omega t = \text{const} \quad \Rightarrow \quad \boxed{v_p = \frac{\omega}{k}}$$

Key properties:

- ▶ Phase velocity can exceed speed of light c
- ▶ No violation of relativity: no information transfer at v_p
- ▶ In dispersive media, v_p depends on wavelength

Defining group velocity

Derivation from two-wave superposition:

Consider two waves with slightly different frequencies:

$$\psi_1 = \cos(kz - \omega t) \quad (7)$$

$$\psi_2 = \cos((k + dk)z - (\omega + d\omega)t) \quad (8)$$

Their sum creates a beat pattern (envelope):

$$\psi_1 + \psi_2 = 2 \cos\left(\frac{dk \cdot z - d\omega \cdot t}{2}\right) \cos(kz - \omega t)$$

Envelope moves where its argument is constant:

$$v_g = \frac{\partial \omega}{\partial k}$$

Group velocity is the derivative of the dispersion relation $\omega(k)$

Group velocity carries energy and information

Physical significance of v_g :

- ▶ Energy propagates in the wave packet *envelope*
- ▶ Information travels with the packet shape at v_g
- ▶ Individual crests passing through envelope carry no net energy

For electrons:

- ▶ Group velocity is equivalent to classical particle velocity
- ▶ $v_g = \partial\omega/\partial k$ connects wave and particle pictures

Common misconceptions in the uncertainty principle

Misconception 1: Uncertainty comes from measurement error

In reality: Uncertainty is **intrinsic** to quantum reality, not instrumental limitation

Misconception 2: phase velocity is the particle velocity

In reality: $v_p \neq$ particle velocity; **group velocity** v_g carries energy

Misconception 3: “sometimes wave, sometimes particle”

In reality: Electrons exhibit simultaneous wave and particle behavior

Summary: the foundation of wave-like behavior for electrons

Three pillars of electron-wave

$$\boxed{\lambda = \frac{h}{p}, \quad \Delta p \cdot \Delta z = h, \quad v_g = \frac{\partial \omega}{\partial k}} \quad (9)$$

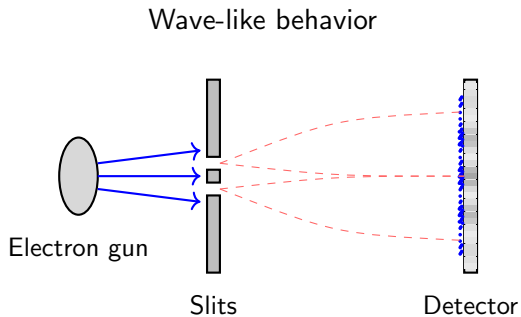
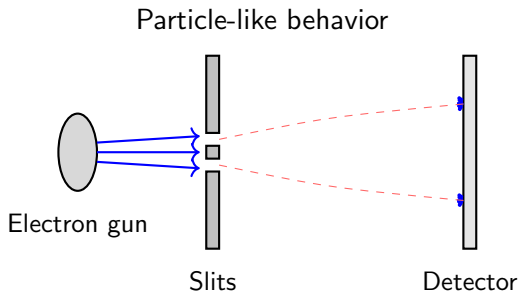
- ▶ **de Broglie relation** $\lambda = h/p$: bridges particle momentum to wave interference
- ▶ **Uncertainty principle** $\Delta p \cdot \Delta z = h$: sets fundamental limits on localization
- ▶ **Group velocity** $v_g = \partial \omega / \partial k$: connects wave packets to electron dynamics

These relations form the foundation for our course of physics of materials

Pause

Electron as a wave: we have now seen that electrons behave as waves AND particles.
Would it be enough to use the wave equation to describe the electron?

Wave-particle duality upon observations



Introducing the wave function Ψ

- ▶ Electron described by complex wave function $\Psi(\mathbf{r}, t)$ at every point in space
- ▶ Ψ itself is not directly observable—it is a mathematical construct
- ▶ Contains all information about the quantum state of the electron

Wave Function

$$\Psi(\mathbf{r}, t) : \mathbb{R}^4 \rightarrow \mathbb{C}$$

Notation:

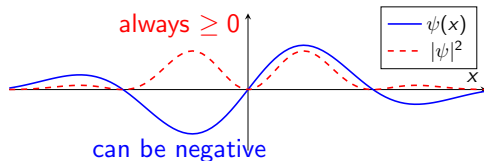
- ▶ $\mathbf{r}$ = position vector (x, y, z)
- ▶ t = time
- ▶ $\mathbb{C}$ = complex numbers (amplitude and phase)

The probabilistic interpretation: Born rule

- ▶ $|\Psi|^2$ gives probability density of finding electron at position $\mathbf{r}$
- ▶ Probability in volume dV equals $|\Psi|^2 dV$ —always real and non-negative
- ▶ Wave function must be normalized: total probability equals one

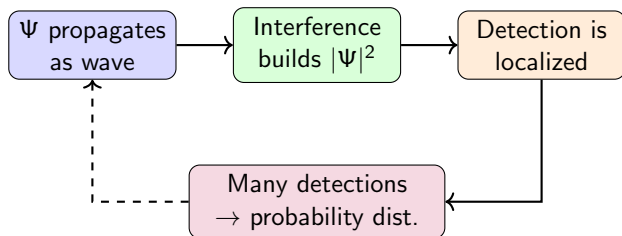
Born Rule

$$P(\mathbf{r}, t) = |\Psi(\mathbf{r}, t)|^2 = \Psi^* \Psi$$



Resolving wave-particle duality

- ▶ Wave function propagates as wave—explains interference and diffraction
- ▶ Measurement collapses to particle-like detection at specific location
- ▶ Many measurements reproduce $|\psi|^2$ distribution: probabilistic predictions



Checkpoint: wave function interpretation

- ▶ Why is $|\psi|^2$ observable but ψ itself is not directly measurable?
- ▶ How does the probabilistic interpretation resolve wave-particle duality?
- ▶ What physical constraint requires the wave function to be normalized?

Normalization Condition

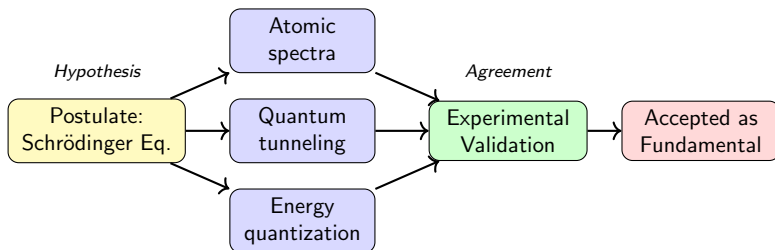
$$\int_{\text{all space}} |\Psi|^2 dV = 1$$

Key insight: The electron must be *somewhere*—total probability = 1.

Think about: What happens to $|\psi|^2$ if we multiply ψ by a phase factor $e^{i\phi}$?

The Schrödinger equation: a fundamental postulate

- ▶ Schrödinger equation is **NOT** derived from more fundamental principles
- ▶ It is a *postulate*—a brilliant hypothesis validated by experimental agreement
- ▶ Similar to Newton's laws: accepted because predictions match observations



The time-dependent Schrödinger equation

- ▶ Governs evolution of quantum state $\Psi(\mathbf{r}, t)$ in space and time
- ▶ Complex-valued wave function: both amplitude and phase carry information
- ▶ First-order in time, second-order in space (unlike classical wave equation)

Time-Dependent Schrödinger Equation

$$-\frac{\hbar^2}{2m}\nabla^2\Psi + V\Psi = i\hbar\frac{\partial\Psi}{\partial t}$$

Left side: Kinetic + Potential energy acting on Ψ

Right side: Time evolution of Ψ

Notation: key symbols in Schrödinger equation

- ▶ $\hbar = h/2\pi \approx 1.055 \times 10^{-34} \text{ J}\cdot\text{s}$ (reduced Planck constant)
- ▶ m = electron mass (or effective mass in solids)
- ▶ $V = V(\mathbf{r}, t)$ potential energy function (environment)
- ▶ $\Psi = \Psi(\mathbf{r}, t)$ complex wave function; $|\Psi|^2 dV$ = probability in dV

The Laplacian Operator

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Physical meaning: $-\frac{\hbar^2}{2m}\nabla^2\Psi$ represents kinetic energy operator acting on Ψ

Separating space and time in Schrödinger equation

- ▶ **Goal:** Find solutions with definite energy from time-dependent equation
- ▶ Assume separable form: $\Psi(\mathbf{r}, t) = \psi(\mathbf{r}) \cdot T(t)$
- ▶ Separation works when potential V is time-independent

Separable Ansatz

$$\Psi(\mathbf{r}, t) = \psi(\mathbf{r}) \cdot T(t)$$

Key insight: Substituting into Schrödinger equation:

$$\underbrace{\frac{1}{\psi} \left[-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi \right]}_{\text{depends only on } \mathbf{r}} = \underbrace{\frac{i\hbar}{T} \frac{dT}{dt}}_{\text{depends only on } t} = E$$

Each side must equal the *same constant* E (the energy eigenvalue)

Separation of variables: the derivation

- ▶ Substitute $\Psi = \psi T$ into time-dependent Schrödinger equation
- ▶ Divide both sides by ψT to isolate spatial and temporal parts
- ▶ Left side depends only on $\mathbf{r}$; right side depends only on t
- ▶ Both sides must equal separation constant E (the energy)

Separated form:

$$\frac{1}{\psi} \left[-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi \right] = \frac{i\hbar}{T} \frac{dT}{dt} = E \quad (10)$$

Time-independent Schrödinger equation

- ▶ Spatial part yields eigenvalue equation: operator acting on ψ gives $E \cdot \psi$
- ▶ E is the energy eigenvalue—a measurable quantity
- ▶ Solutions $\psi(\mathbf{r})$ are energy eigenfunctions with definite energy E
- ▶ This is the central equation for finding allowed quantum states

Time-Independent Schrödinger Equation

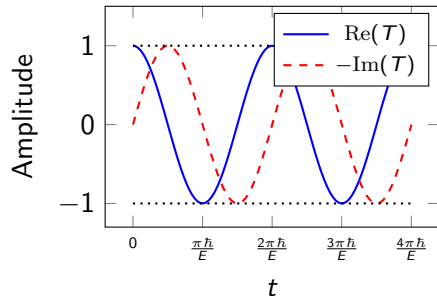
$$\boxed{-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi} \quad (11)$$

Notation:

- ▶ $\psi(\mathbf{r})$: spatial wave function
- ▶ E : energy eigenvalue (constant)
- ▶ V : potential energy

Time evolution: the oscillating phase factor

- ▶ Temporal equation: $i\hbar \frac{dT}{dt} = ET$ has exponential solution
- ▶ Time dependence is purely oscillatory phase factor
- ▶ Full solution: $\Psi(\mathbf{r}, t) = \psi(\mathbf{r}) \cdot e^{-iEt/\hbar}$
- ▶ Probability density $|\Psi|^2 = |\psi|^2$ is time-independent!

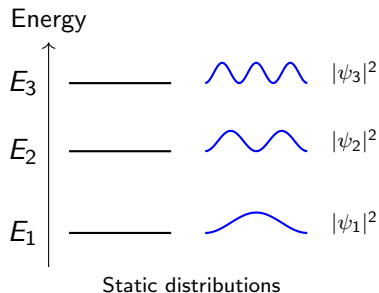


Time evolution factor:

$$T(t) = e^{-iEt/\hbar} \quad (12)$$

Stationary states: why the name?

- ▶ Stationary states have time-independent probability distributions
- ▶ Energy E is precisely defined—no uncertainty in energy measurement
- ▶ All expectation values $\langle A \rangle$ are constant in time
- ▶ General solutions: superpositions of stationary states with different E



Probability density:

$$|\Psi(\mathbf{r}, t)|^2 = |\psi(\mathbf{r})|^2 \quad (13)$$

Expectation values: quantum averages

Quantum mechanics predicts average outcomes over many measurements.

Expectation Value Definition

$$\langle A \rangle = \frac{\int_V \psi^* \hat{A} \psi dV}{\int_V |\psi|^2 dV} \quad (14)$$

Notation:

- ▶ $\hat{A}$ = quantum operator for observable A
- ▶ For normalized ψ : denominator = 1

Common examples:

$$\langle z \rangle = \int \psi^* z \psi dV \quad (\text{average position}) \quad (15)$$

$$\langle p \rangle = \int \psi^* (-i\hbar \nabla) \psi dV \quad (\text{average momentum}) \quad (16)$$

Pause

Given the Schrödinger's equation, now we can compute a few examples on what happens with the electron under multiple assumptions. In particular, we will look at what happens if we place the electrons in different potentials

Free electron wave function ($V = 0$)

- ▶ **Single region (all z):** $V(x) = 0 \Rightarrow$ no interface, no scattering, no reflection
- ▶ **Plane-wave solution:**

$$\psi(x, t) = A e^{i(kx - \omega t)}$$

- ▶ **Stationary-state form (spatial part):**

$$\psi(z) = A e^{ikx}$$

- ▶ General free-particle solution is a superposition of right- and left-moving waves,

$$\psi(z) = A e^{ikx} + B e^{-ikx},$$

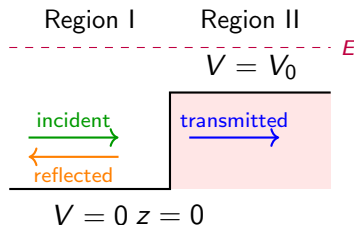
but for a *single* plane wave traveling to the right we take $B = 0$.

Dispersion and wave vector

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad \omega = \frac{E}{\hbar} = \frac{\hbar k^2}{2m} \quad (17)$$

Potential step: setup and classical expectation

- ▶ Electron with energy E approaches potential step V_0 at $z = 0$
- ▶ **Classical:** full transmission if $E > V_0$, full reflection if $E < V_0$
- ▶ **Quantum:** wave nature creates partial reflection even when $E > V_0$
- ▶ Must solve Schrödinger equation in each region and match at boundary



Wave functions in each region ($E > V_0$)

- ▶ **Region I** ($z < 0$): $\psi_1 = Ae^{ik_1z} + Be^{-ik_1z}$ with $k_1^2 = 2mE/\hbar^2$
- ▶ **Region II** ($z > 0$): $\psi_2 = Ce^{ik_2z}$ with $k_2^2 = 2m(E - V_0)/\hbar^2$
- ▶ No leftward wave in Region II (no source at $z = +\infty$)
- ▶ A = incident amplitude, B = reflected, C = transmitted

Wave Vectors in Each Region

$$k_1 = \sqrt{\frac{2mE}{\hbar^2}}, \quad k_2 = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} \quad (18)$$

Note: $k_2 < k_1$ when $E > V_0$ (slower wave in Region II)

Boundary conditions at $z = 0$

- ▶ **Continuity of ψ :** $\psi_1(0) = \psi_2(0) \rightarrow A + B = C$
- ▶ **Continuity of $d\psi/dz$:** ensures probability current conservation
- ▶ At $z = 0$: $ik_1(A - B) = ik_2C$
- ▶ Two equations, three unknowns $\rightarrow$ express B and C in terms of A

Boundary Conditions at Interface

$$A + B = C \quad (19)$$

$$k_1(A - B) = k_2C \quad (20)$$

Physical meaning: Wave function and its slope must be continuous for finite potential steps.

Reflection and transmission coefficients

- ▶ Solve boundary equations: $\frac{B}{A} = \frac{k_1 - k_2}{k_1 + k_2}$, $\frac{C}{A} = \frac{2k_1}{k_1 + k_2}$
- ▶ Reflection coefficient $R = |B/A|^2$ measures reflected probability
- ▶ Transmission coefficient $T = (k_2/k_1)|C/A|^2$ accounts for velocity change
- ▶ Verify $R + T = 1$: probability conservation satisfied

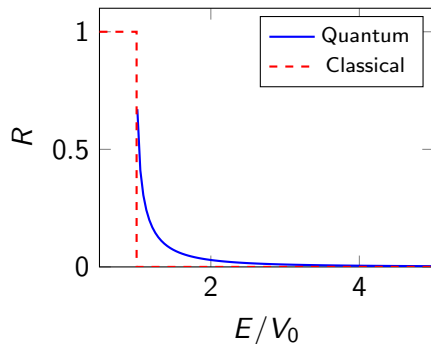
Reflection and Transmission Coefficients

$$R = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2, \quad T = \frac{4k_1 k_2}{(k_1 + k_2)^2} \quad (21)$$

Check: $R + T = \frac{(k_1 - k_2)^2 + 4k_1 k_2}{(k_1 + k_2)^2} = \frac{(k_1 + k_2)^2}{(k_1 + k_2)^2} = 1 \quad \checkmark$

Quantum vs classical: partial reflection

- ▶ Even when $E > V_0$, quantum mechanics predicts $R \neq 0$
- ▶ Wave impedance mismatch causes reflection (like light at glass interface)
- ▶ $R \rightarrow 0$ only when $k_1 = k_2$ (no step) or $E \gg V_0$
- ▶ Fundamentally different from classical all-or-nothing behavior



Checkpoint: boundary condition application

Conceptual Questions

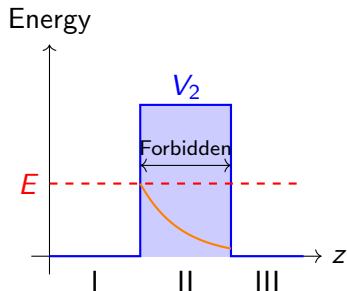
1. Why must both ψ and $d\psi/dz$ be continuous at the boundary?
2. What happens to R as E approaches V_0 from above?
3. Why is there no $D \cdot e^{-ik_2z}$ term in Region II?
4. How does this result connect to optical reflection at interfaces?

Hint for Q4: Compare $R = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2$ with Fresnel equations for normal incidence:

$$R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

Classically forbidden regions: when $E < V$

- ▶ Classical particle cannot exist where kinetic energy would be negative ($E < V$)
- ▶ Quantum mechanics: wave function doesn't vanish—it **decays exponentially**
- ▶ Region where $E < V_2$ is called the **classically forbidden region**
- ▶ Probability density $|\psi|^2$ remains finite inside barrier



Wave function inside the barrier

Schrödinger equation inside barrier:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dz^2} + V_2\psi = E\psi \quad (22)$$

Rearranging:

$$\frac{d^2\psi}{dz^2} = \frac{2m}{\hbar^2}(V_2 - E)\psi = \gamma^2\psi \quad \text{where } \gamma^2 > 0 \quad (23)$$

Decay Constant in Barrier

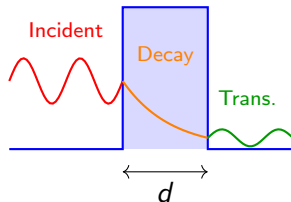
$$\gamma = \sqrt{\frac{2m(V_2 - E)}{\hbar^2}} \quad (24)$$

Solution: Real exponentials (not oscillatory)

$$\psi_2 = Ae^{-\gamma z} + Be^{+\gamma z} \quad (25)$$

Exponential decay through finite barrier

- ▶ For barrier of width d : keep both exponential terms inside
- ▶ Wave function decays as $e^{-\gamma z}$ through barrier
- ▶ Amplitude at exit reduced by factor $e^{-\gamma d}$
- ▶ Probability reduced by $e^{-2\gamma d}$



Amplitude Decay

$$\psi_2(z) \propto e^{-\gamma z}, \quad |\psi|^2 \propto e^{-2\gamma d}$$

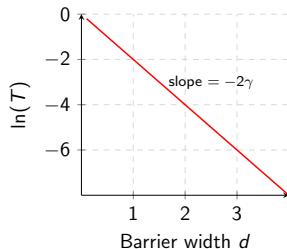
Tunneling transmission probability

- ▶ Apply boundary conditions at both interfaces ($z = 0$ and $z = d$)
- ▶ Transmission coefficient: $T = \frac{|\text{transmitted}|^2}{|\text{incident}|^2}$
- ▶ For thick barriers ($\gamma d \gg 1$): $T \approx T_0 e^{-2\gamma d}$

Tunneling Probability (Thick Barrier)

$$T \approx T_0 \exp\left(-\frac{2d}{\hbar} \sqrt{2m(V_2 - E)}\right)$$

Doubling barrier width squares the exponential suppression



Physical interpretation: energy-time uncertainty

Energy-Time Uncertainty Relation

$$\Delta E \cdot \Delta t \geq \hbar \quad (26)$$

- ▶ Tunneling doesn't violate energy conservation—uses uncertainty principle
- ▶ Electron “borrows” energy $\Delta E = V_2 - E$ for time $\Delta t \approx d/v$
- ▶ Faster traversal (thinner barrier) allows larger energy borrowing
- ▶ Heavier particles tunnel less: $\gamma \propto \sqrt{m}$ means exponentially suppressed

Key insight: Quantum mechanics allows temporary energy fluctuations within the bounds set by the uncertainty principle

Calculating uncertainties from expectation values

Uncertainty = root-mean-square deviation from average:

Uncertainty Formula

$$\Delta z = \sqrt{\langle z^2 \rangle - \langle z \rangle^2} \quad (27)$$

Procedure:

1. Calculate $\langle z \rangle = \int \psi^* z \psi dV$
2. Calculate $\langle z^2 \rangle = \int \psi^* z^2 \psi dV$
3. Compute $\Delta z = \sqrt{\langle z^2 \rangle - \langle z \rangle^2}$

Same for momentum:

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} \quad (28)$$

Product $\Delta z \Delta p \geq \hbar$ always satisfied for any valid ψ .

Chapter summary: foundations of quantum physics

Key takeaways:

- ▶ Schrödinger equation is fundamental **postulate**
- ▶ Boundary conditions $\rightarrow$ quantization & tunneling
- ▶ Uncertainty principle sets fundamental limits
- ▶ Expectation values connect formalism to measurements